

Birla Institute of Technology & Science, Pilani
First Semester 2025-2026 (Mid Semester Examination)
ECE F314: Electromagnetic Field & Microwave Engineering
ECE F331: Microwave Engineering
(Closed Book)

Part: A

Duration: 45 minutes

Total Marks: 30

Note to Students:

Please write the answers in the answer sheet provided with proper units and signs. Marks will be deducted for improper signs and units.

1. The reflection coefficient at a short-circuited end of a transmission line is..... [1]
2. On a Smith Chart, a movement clockwise corresponds to moving towards the [1]
source/generator
3. The input impedance of a $\lambda/4$ (quarter-wavelength) transmission line terminated with an open circuit is [1]
0
4. An unpolarized wave is incident from air into paraffin ($\epsilon_r = 3$). Find the incident angle for which the reflected wave is completely TE polarized. [2]

$$\theta = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \tan^{-1} \sqrt{3} = 60^\circ$$

5. A conductor is characterized by a conductivity, $\sigma = 100 \text{ S/m}$, and permittivity $\epsilon = 4 \epsilon_0$. At what angular frequency ω does the amplitude of the conduction current density equal the amplitude of the displacement current density? [2]

$$6R = \omega \epsilon \epsilon_r R$$

$$\omega = \frac{6}{\epsilon_0 \epsilon_r} = \frac{2.82 \times 10^{12}}{\text{rad/s}}$$

6. Write down Maxwell's equation, which is based on Faraday's law (Both integral and differential form). [2]

$$\textcircled{1} \rightarrow \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -j\omega \mu \vec{H} \quad \text{Differential form}$$

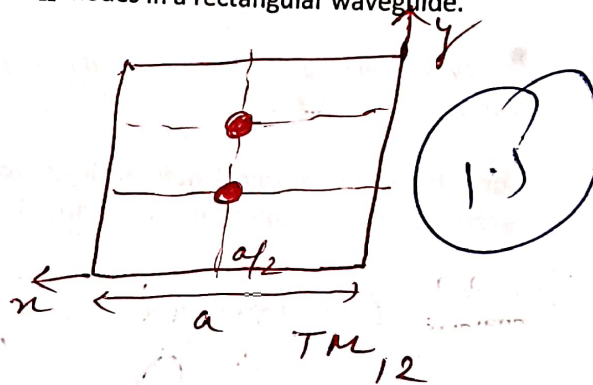
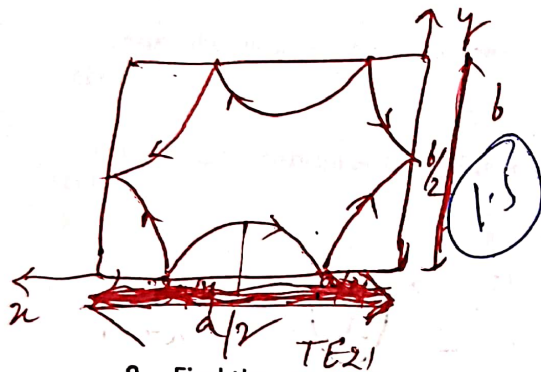
$$\textcircled{1} \rightarrow \oint \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \int \vec{B} \cdot d\vec{s} \quad \text{Integral form}$$

7. A rectangular waveguide filled with a dielectric of $\epsilon_r = 4$, with a cross-section of 5 cm * 2 cm, propagates the TM_{11} mode at 9 GHz. Find the cut-off wavelength [2]

$$f_c = \frac{c}{2 \times \sqrt{\epsilon_r} \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2}} = 4.03 \text{ GHz}$$

$$\lambda_c = \frac{c}{2 \times f_c} = 3.7 \text{ cm}$$

8. Draw the field pattern for TE_{21} and TM_{12} modes in a rectangular waveguide. [3]



9. Find the ratio of free space wavelength to wavelength in a conductor with $\sigma = 10^4 \text{ S/m}$ and $\mu_r = 15$ at 10 KHz. [3]

$$\lambda_0 = \frac{c}{f} = \sqrt{3 \times 10^4} \text{ m} \quad \text{①}$$

$$\lambda_c = \frac{2\pi}{\beta} = \sqrt{0.0817} \text{ m} \quad \text{①} \quad \beta = \sqrt{\frac{\mu \omega \sigma}{2}}$$

$$\frac{\lambda_0}{\lambda_c} = \frac{3 \times 10^4}{0.0817} = \boxed{3.67 \times 10^5} \quad \text{①}$$

10. A lossless transmission line has a characteristic impedance of 50Ω and is terminated in a load of 100Ω . The magnitude of a voltage wave incident on the line is 20 V (rms) . Determine the maximum voltage and minimum current on the line. [4]

$$\Gamma = \frac{100 - 50}{100 + 50} = \frac{1}{3}$$

$$V_{inc} = 20\text{ V}_{rms}$$

$$I_{inc} = \frac{20}{50} = 0.4\text{ A}_{rms}$$

$$V_{max} = 20 \left(1 + \frac{1}{3}\right) = \frac{80}{3} = 26.67\text{ V}_{rms}$$

$$I_{min} = 0.4 \left(1 - \frac{1}{3}\right) = 0.2667\text{ A}_{rms}$$

11. A uniform plane wave $\vec{E} = E_0 \hat{a}_y$ propagates in the x direction in a simple lossless medium characterized by $\epsilon_r = 9$, $\sigma = 0$ and $\mu_r = 4$. E_y is sinusoidal and has a frequency of 100 MHz . Its maximum value is 0.01 V/m at $t = 0$ and $x = 1/6$ meter. Obtain an instantaneous expression of $\vec{H}$ in terms of t and x . [4]

$$\beta = \omega \sqrt{\mu_r \epsilon_r \epsilon_0 \mu_0}$$

$$\eta = 377 \times \frac{2}{3} = 251.33\Omega$$

$$\beta = 4\pi\text{ rad/m}$$

$$E_y \Big|_{t=0, x=1/6} = 0.01 = E_0 \sin(\omega t - \beta x + \phi)$$

$$E_0 \sin\left(-\frac{2\pi}{3} + \phi\right) = 0.01$$

$$-\frac{2\pi}{3} + \phi = \pi/2$$

$$\phi = \frac{7\pi}{6}$$

$$\vec{E} = 0.01 \sin\left(2\pi \times 10^8 t - 4\pi x + \frac{7\pi}{6}\right) \hat{a}_y\text{ V/m}$$

$$\vec{H} = \left[\frac{0.01}{251.33}\right] \sin\left(2\pi \times 10^8 t - 4\pi x + \frac{7\pi}{6}\right) \hat{a}_z\text{ A/m}$$

$$\vec{H} = 3.97 \times 10^{-5} \sin\left(2\pi \times 10^8 t - 4\pi x + \frac{7\pi}{6}\right) \hat{a}_z\text{ A/m}$$

$$\vec{H} = 3.97 \times 10^{-5} \cos\left(2\pi \times 10^8 t - 4\pi x + \frac{2\pi}{3}\right) \hat{a}_z\text{ A/m}$$

if $\cos$ which is considered

$$\phi = \frac{2\pi}{3}$$

12. In free space $Z \leq 0$, a plane wave has $H = 12 \cos(18 \times 10^7 t - \beta z) \hat{a}_x$ (mA/m). The wave is incident normally on a lossless medium characterized by $\epsilon_r = 9$, and $\mu_r = 4$ in region $Z \geq 0$. Obtain reflected wave (E_r, H_r) and transmitted wave (E_t, H_t). [5]

Normal
incidence

$$\eta_1 = 377 \Omega = 120 \pi$$

$$\eta_2 = 251.33 = 80 \pi$$

$$\Gamma = \frac{-40 \pi}{2 \omega \pi} = -0.2$$

$$\tau = 1 + \Gamma = 0.8$$

$$H_i = 12 \cos(18 \times 10^7 t - \beta z) \hat{a}_x \text{ mA/m}$$

$$|E_i| = 12 \times 377 \times 10^{-3} = 4.524 \text{ V/m}$$

$$|E_r| = 0.2 \times 4.524 = 0.905 \text{ V/m}$$

$$|E_t| = 0.8 \times 4.524 = 3.619 \text{ V/m}$$

$$\beta_1 = 0.6 \text{ rad/m}$$

$$\beta_2 = 3.6 \text{ rad/m}$$

$$|H_r| = \frac{0.905}{377} = 2.4 \text{ mA/m}$$

$$|H_t| = \frac{3.619}{251.33} = 14.4 \text{ mA/m}$$

$$E_i = -1.44 \pi \cos(18 \times 10^7 t - 0.6 z) \hat{a}_y \text{ V/m}$$

$$E_r = -0.905 \cos(18 \times 10^7 t + 0.6 z) \hat{a}_y \text{ V/m}$$

$$H_r = -2.4 \cos(18 \times 10^7 t + 0.6 z) \hat{a}_x \text{ mA/m}$$

$$E_t = -3.619 \cos(18 \times 10^7 t - 3.6 z) \hat{a}_y \text{ V/m}$$

$$H_t = 14.4 \cos(18 \times 10^7 t - 3.6 z) \hat{a}_x \text{ mA/m}$$

$$H_i = 12 \cos(18 \times 10^7 t - \beta z) \hat{a}_x \text{ (mA/m)}$$

Solutions

15. $\sigma = 0.4 \times 10^{-2} \text{ S/m}$

$$\omega = 2\pi f = 25 \times 10^6 \text{ Hz}$$

$$\epsilon_r = 15$$

$\therefore$ let's check, $\frac{\sigma}{\omega \epsilon}$

$$0.4 \times 10^{-2}$$

$$2 \times 25 \times 10^6 \times 15 \times 8.85 \times 10^{-12}$$

$$= 2.39 \approx 2.4$$

Since $\frac{\sigma}{\omega \epsilon} > 1$ therefore the

material is conductive in nature.

$$\text{Hence, } \alpha = \beta = \sqrt{\pi f \mu \sigma}$$

$$\alpha = 0.177 \text{ Np/m}$$

$$\beta = 0.177 \text{ rad/m}$$

$$N_p = \frac{\omega}{\beta} = \frac{25 \times 10^6 \times 2}{0.177} = 7.07 \times 10^8$$

Q2:- $\rho_s = 4 \text{ nC/m}^2$.

(A) for point (4, -2, 1) inside the conductor

② | $y > 0, \epsilon_1 = 3$
 ① | $y < 0$

$\therefore E = 0 = D$

(B) for point (-3, 1, 4) which is inside dielectric medium

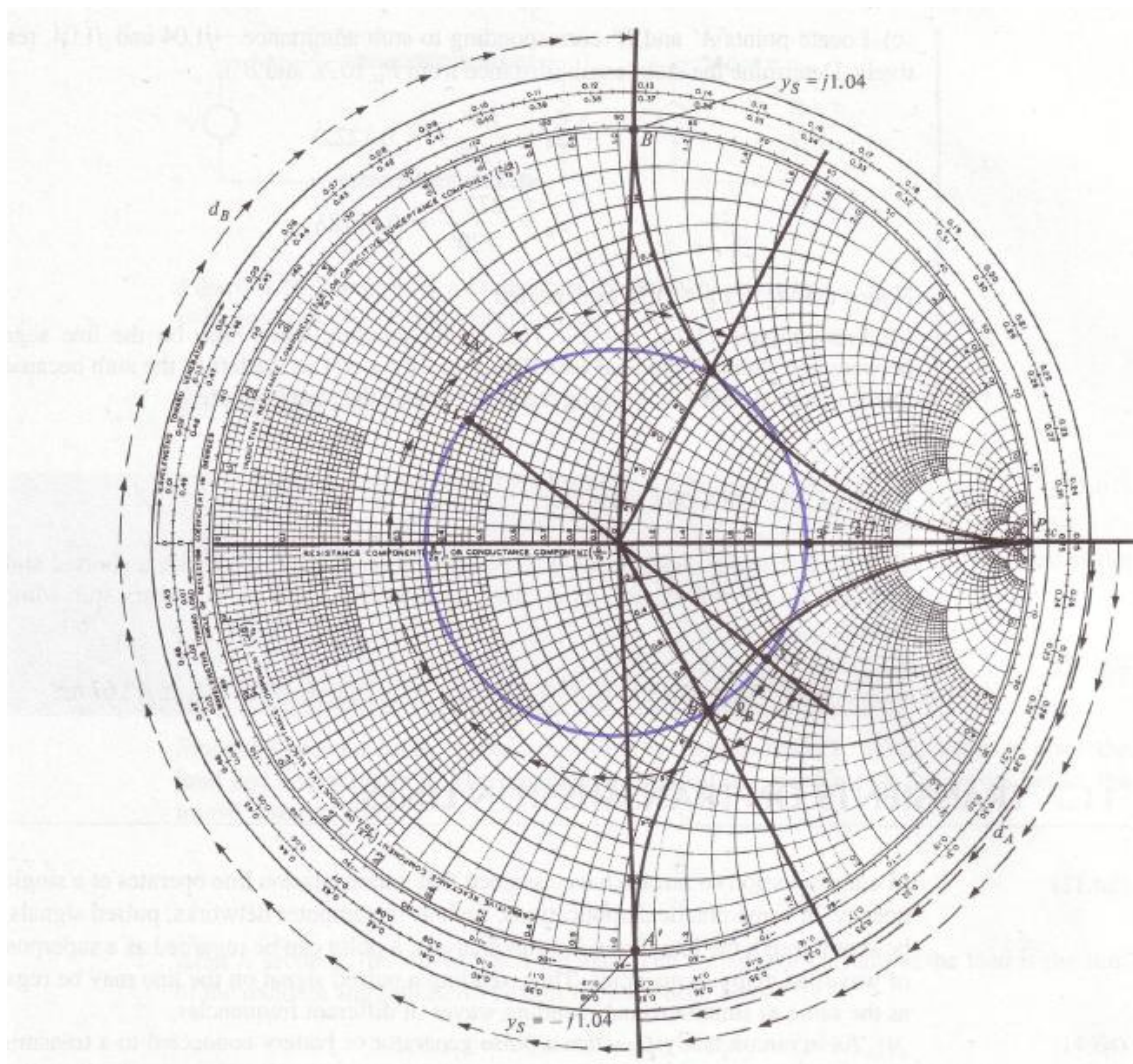
$E_t = 0$ | for conductor, $\therefore E_{t2} = 0$

and, $D_n = \rho_s = 4 \text{ nC/m}^2$ | $D_{1n} = D_{2n}$
 ① $\therefore D_{1n} = 4 \text{ nC/m}^2 = D_{2n} = D_2$

and, $E_{1n} = \frac{D_1}{\epsilon_0 \epsilon_1} = \frac{4 \times 10^{-9}}{8.85 \times 10^{-12} \times 3} = 150.66 \text{ V/m}$



3.



Solution:

$$(a) z_L = \frac{Z_L}{Z_0} = \frac{40 + j30}{100} = 0.4 + j0.3$$

Locate z_L on the Smith chart as in Figure 11.24 and from this draw the s -circle so that y_L can be located diametrically opposite z_L . Thus $y_L = 1.6 - j1.2$. Alternatively, we may find y_L using

$$y_L = \frac{Z_0}{Z_L} = \frac{100}{40 + j30} = 1.6 - j1.2$$

Locate points A and B where the s -circle intersects the $g = 1$ circle. At A, $y_s = -j1.04$ and at B, $y_s = +j1.04$. Thus the required stub admittance is

$$Y_s = Y_0 y_s = \pm j1.04 \frac{1}{100} = \pm j10.4 \text{ mS}$$

Both $j10.4 \text{ mS}$ and $-j10.4 \text{ mS}$ are possible values.

(c) Locate points A' and B' corresponding to stub admittance $-j1.04$ and $j1.04$, respectively. Determine the stub length (distance from P_{sc} to A' and B'):

$$d_A = \frac{88^\circ}{720^\circ} \lambda = 0.1222\lambda$$

$$d_B = \frac{272^\circ}{720^\circ} \lambda = 0.3778\lambda$$

Notice that $d_A + d_B = 0.5\lambda$ as expected.

(b) From Figure 11.24, we determine the distance between the load (antenna in this case) y_L and the stub. At A ,

$$\ell_A = \frac{\lambda}{2} - \frac{(62^\circ - -39^\circ)\lambda}{720^\circ} = 0.36\lambda$$

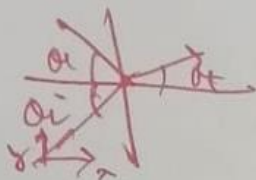
At B :

$$\ell_B = \frac{(62^\circ - 39^\circ)}{720^\circ} \lambda = 0.032\lambda$$

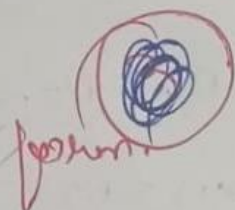
Q4:- $E = 8 \cos(\omega t - 4x - 3z)$ ay $\sqrt{1}$ m,
 $(z \geq 0)$ $[R = \sqrt{4^2 + 3^2} \Rightarrow 5 = \frac{\omega}{v} \therefore \omega = 3 \times 10^8$
 $f = 15 \times 10^7$ Hz]

(a) Perpendicular (as E-field is along 'y' & wave prop. is along $\hat{x} \hat{z}$)

(b) $\tan \theta = \frac{4}{3} \quad \theta = 53.13^\circ$ (2)



(c) $B_y = B_0 e^{-jRz} (a \sin \theta_i - z \cos \theta_i)$ ay
 reflected E-field in phase



$B_r = B_i$

$\therefore B_r = (4x - 3z) \cdot \text{rad/m}$

$(4\hat{x} + 3\hat{z}) \cdot (4\hat{x} + 3\hat{z}) = (4^2 + 3^2) = 25$
 $(4\hat{x} - 3\hat{z}) \cdot (4\hat{x} + 3\hat{z}) = 16 - 9 = 7$

also, $\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$

Contd.

$\eta_2 = 238.4 \Omega, \eta_0 = \eta_1 = 377 \Omega$

$\therefore \Gamma_{\perp} = -0.389$ (1)

$E_{r0} = -0.389 \times 8 = -3.112 \text{ V/m}$

$$\therefore E_1 = -3.112 \cos(\omega t - 4x + 3z) \underline{a_y \text{ V/m}}$$

(2)

(d) $H_{ts} = \frac{E_{to}}{\eta_2} (-\cos \theta_t a_x + \sin \theta_t a_z) e^{-j\beta_2 (x \sin \theta_t + z \cos \theta_t)}$

(3) from Snell's law

write
power
in

(1)

$$\theta_t = 30.34^\circ$$

$$\text{Now, } \eta_2 = 238.4 \Omega$$

$$\left[\eta_1 \sin \theta_i = \eta_2 \sin \theta_t \right]$$

$$\beta_2 = \omega \sqrt{\mu_0 \epsilon_2}$$

$$= \frac{1.5 \times 10^8}{3 \times 10^8} \times \sqrt{2.5}$$

$$= 7.906$$

$$\tau = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

$$= \frac{0.611}{1}$$

$$\therefore E_{to} = 0.611 \times 8 = \underline{4.88 \text{ V/m}}$$

$$H_{ts} = \frac{4.88}{238.4} (-\cos 30.34 a_x + \sin 30.34 a_z)$$

(phasor form).

$$e^{-j7.906 (x \sin 30.34^\circ + z \cos 30.34^\circ)}$$

(1)

$$H_t = (-17.7 a_x + 10.4 a_z)$$

$$\cos(\omega t - 4x - 6.9z) \underline{\underline{\text{mA/m}}}$$

